

0 6

The human papilloma virus contains a double-stranded DNA genome.

0 6 . 1

Which components are found in a human papilloma virus?

Tick (✓) **one** box.

[1 mark]

Capsid and attachment protein

☐

Capsid, attachment protein and reverse transcriptase

☐

Capsule and attachment protein

☐

Cell-surface membrane and attachment protein

☐

0 6 . 2

The DNA-replication enzymes of a human cell make copies of the human papilloma virus genome.

Name **two** enzymes that are involved in replicating the DNA of the human papilloma virus and describe their roles in the replication process.

[3 marks]

Name of enzyme 1 _____

Role of enzyme 1 _____

Name of enzyme 2 _____

Role of enzyme 2 _____

Question 6 continues on the next page

Turn over ►



06.3

Human papilloma virus infects cells that are no longer dividing. The human papilloma virus genome contains genes that code for proteins that cause human cells to restart their cell cycles.

Human papilloma virus infection can cause cancer.

Explain why.

[1 mark]

06.4

Human papilloma virus (HPV) is transmitted through sexual contact.

More than 95% of cervical cancers (which only affect females) are due to HPV infection. HPV infection of other tissues increases the risk of cancer but this is rare compared with cervical infection.

A vaccine is available that is over 80% effective at preventing HPV infection, if given before the person has been exposed to HPV. There is evidence of herd immunity when more than 50% of the population have been vaccinated.

Evaluate whether 10- to 12-year-old boys should be given the HPV vaccine.

[4 marks]

